nist wellness collective Body Politic. It was later found that a quarter of the respondents claiming to have persistent symptoms had not tested positive for Covid, and half had never been tested. There have been more reliable studies since then, involving only Covid-positive respondents and published in peer-reviewed journals.

In the early days of the pandemic, some speculated that the virus might be entering the brain and infecting neurons, the cells responsible for cognitive processing. But recent studies show that the blood-brain barrier is the furthest the virus gets—it doesn't directly attack neurons. A July 2022 study by the National Institutes of Health in the US found no evidence of the virus directly entering the brain. "Inflammation, reduction of blood flow or a clot travelling to the brain through blood are more commonly seen as reasons for neurological damage," says Dr Jyoti Sehgal, associate director, neurology, at the Institute of Neurosciences, Medanta – The Medicity, Gurugram. A study of people who died due to Covid, published in January 2022 by University of California neurologists, found that in 66 per cent of the cases, the virus had affected brain cells known as astrocytes. This could be the reason for symptoms such as fatigue, depression and brain fog. Evidence from University College London has now suggested a second way in which Covid affects the brain—by blocking blood flow to the organ. This happens when receptors called pericytes are damaged, causing tissues to constrict. Reduced blood flow can impair the functioning of neurons and even kill them.

Evidence has also emerged for a third path through which Covid can impact the brain. Research by the US-based National Institute of Neurological Disorders and Stroke shows that inflammation results in the immune system attacking and even damaging brain cells. Antibodies—proteins produced by the immune system in response to Covid—have been shown to damage the cells forming the blood-brain barrier. "Damage to this barrier can cause bleeding and clots in patients, increasing the risk of stroke," says Dr Sehgal. "Diagnosing how Covid has affected your body and brain will determine the kind of treatment. If one finds clots or reduction in blood flow, then the condition is more serious than when you have symptoms of brain fog with no physical damage to the organ."



"Damage to the blood-brain barrier can cause bleeding and clots in patients, increasing the risk of stroke"

Dr Jyoti Sehgal
Associate Director, Neurology, Institute of Neurosciences, Medanta - The Medicity, Gurugram

HOW TO STAY SAFE

Vaccination helps reduce the chances of brain fog. A self-reported study of 1.2 million UK citizens in November 2021 found that two doses halved the risk of long-Covid symptoms. Analysing eight studies published before mid-January 2022, the UK's Health Security Agency found that six of them showed vaccines reduced post-Covid complications. "It is added protection and limits inflammation, which in turn reduces lasting complications," says Dr Sehgal.

Many doctors advise even those who are vaccinated to get basic blood tests done during their illness as vaccine immunity wanes and those who are elderly or have comorbidities can still be at risk of neurological complications. "I suggest four tests—Interleukin 6 or IL-6, D-dimer, complete blood count and C-reactive protein," says Dr Srivastava. These can help doctors ascertain the need for further tests. "I have observed in my patients that monitoring IL-6 (a chemical associated with inflammation) can reduce chances of post-Covid brain problems," says Dr Renjen. A June 2022 study of 8,077 people published in *PubMed* by German researchers found elevated levels of IL-6 in those with severe long-Covid symptoms. "We need more research on whether continued IL-6 increase leads to brain fog, depression and other brain impairments, but it is a helpful parameter to monitor when you have Covid," adds Dr Renjen.

According to Dr Sehgal, clinical diagnosis of post-Covid health is vital. Depending on symptoms and blood biomarkers, doctors might advise brain imaging, or even a cerebrospinal fluid test, where a needle is inserted to check for abnormalities in brain fluid. "Whether your problems are minor or indicate something major is decided by many factors—age, family history, health status, type of Covid, care during and after infection, and the symptoms," adds Dr Sehgal.

Signs of stroke or worsening of pre-existing neurological diseases such as Alzheimer's or Parkinson's are treated differently from brain fog. "Brain fog can be treated with psychosocial interventions and lifestyle changes, but more serious conditions need medication," says Dr Pawan Ojha, a senior neurologist at Hiranandani Hospital in Vashi, Mumbai.

To cure brain fog, research and medical opinion point to rest, nutrition and mental health support. "Sleeping eight to 10 hours in the night is essential for recovery," says Dr Ojha.